DATA VISUALIZATION / TABLEAU PROJECT

Tourism Trends in India

Dashboard & Story Documentation

Arrivals, destinations, visitor behaviour and tourism revenue

2018-2024	15	5
Growth-table years	States in the extract	Story points

PROJECT AT A GLANCE

Workbook	Tourism Trends India Dashboard
Structure	1 main dashboard | 1 Tableau story
Analysis	5 dashboard KPIs | 6 dashboard charts
Supporting views	COVID score, state-share pie and revenue ranking

Prepared from the supplied packaged Tableau workbook. Findings describe this project extract; they are not official national tourism estimates.

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READING GUIDE

Contents

A complete explanation of the project, its measures and its visual story.

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Five-point chart explanation

Every analytical view is explained through: What do you see? Positive scenario. Negative scenario. Conclusion. Suggestion.

Figure convention: report figures are recreated from the embedded extract, not native Tableau screenshots. Companion views are clearly labelled. All-data totals are used unless a selection is explicitly stated.

01 / PURPOSE

Problem statement and solution

Turn separate tourism tables into a readable analytical story.

Problem statement

Tourism analysis requires several perspectives: where arrivals are concentrated, how recorded activity changes over time, which destinations lead, and how visitor stays relate to revenue. Separate spreadsheets make these comparisons difficult and can hide differences in measurement.

Objectives

- Summarize arrival volume, revenue, average stay and recorded hotel availability.
- Compare state and destination totals in the supplied extract.
- Examine annual arrivals, a moving average and cumulative domestic/foreign totals.
- Explore seasonal volume and the relationship between stay duration and revenue.
- Communicate the results through an interactive dashboard and a five-point story.

Proposed analytical solution

The Tableau workbook combines three related tables. A main dashboard offers headline KPIs and six visualizations; a story adds a weighted COVID-impact score, a state-share view and a revenue ranking. Together, these support exploratory comparisons and a structured classroom presentation.

Scope of the conclusions

The report explains the uploaded project. Dataset provenance, currency and collection methodology are not supplied, so the results should be presented as extract-level findings.

02 / DATASET

Dataset overview and source

Three logical tables packaged in one embedded Hyper extract.

Table	Size	Grain and contents
Tourist Data	200 rows / 7 fields	One row per distinct Tourist_ID; nationality, season, state, arrival volume and nights stayed.
Revenue	200 rows / 6 fields	One row per distinct Tourist_ID; state, revenue, growth category and COVID impact.
Tourism Growth Trends	150 rows / 9 fields	Observation rows containing state, destination, category, year, arrivals, hotels, flights and COVID impact.

Coverage

Characteristic	Verified value
Growth-table years	2018-2024; seven distinct years
Geography	India; 15 state labels in each table
Destination names	15 distinct City_or_Monument values
Visitor category	Domestic and Foreign, in the growth table
Season	Autumn, Monsoon, Summer and Winter, in Tourist Data
Nationality	Eight distinct nationality labels

Source: supplied Tourism_Trends_India_Dashboard(1).twbx. The workbook names an original Excel source, Tourist_Growth_Trends_India_Final. The Excel file and an upstream publication URL are not included. No official provenance is assumed.

Dataset link: https://github.com/moresakshi123/Tableau-Tourist_Growth_Trends_in_India/blob/main/Tourist_Growth_Trends_India_Final.xlsx

03 / PREPARATION

Data quality and preparation

Checks performed on the embedded tables; source values are retained.

Check	Finding / implication
Missing cells	No null cells found in any of the three extracted tables.
Tourist identifiers	200 unique IDs in each visitor/revenue table; matched one-to-one.
State coverage	The same 15 state labels occur in all three tables.
Numeric fields	Arrival volume, nights, revenue, hotels and flights are positive in this extract.
Two arrival measures	Tourist_Arrival_Volume and Tourist_Arrivals have different grains and totals. They must not be treated as interchangeable.
Destination geography	Some state-destination combinations are implausible, such as Hawa Mahal under Tamil Nadu. Geographic assignments need validation.
Decimal arrivals	Some growth-table arrival values are fractional. Their estimation or transformation method is not supplied.
Currency / hotel identity	Currency is unspecified; no hotel ID exists to count distinct active properties.

Preparation workflow

- Extract the packaged data and check field types and unique keys.
- Retain raw values; format display values separately from calculations.
- Use the correct logical table for each measure and state its denominator.
- Treat geography and provenance as limitations until the original source is verified.

04 / RELATIONSHIPS

Data model and analytical grain

Logical relationships preserve table meaning better than a flat many-to-many join.

TOURIST DATA

200 unique Tourist_ID records

Tourist_ID = Tourist_ID | one-to-one in this extract

REVENUE

200 unique Tourist_ID records

Tourist Data.State = Growth Trends.State | repeated states

TOURISM GROWTH TRENDS

150 observations across seven years

Both relationships originate from Tourist Data. Revenue is related by Tourist_ID; Tourism Growth Trends is related by State. There is no direct Year-to-tourist or Category-to-tourist key.

Why this matters

- Revenue and nights can be compared by Tourist_ID or state.
- Year and visitor category belong to the growth table. Combining them with tourist-level measures can repeat state-linked populations.
- A physical join on State could multiply rows. Validate aggregates at their original grain before interpreting cross-table charts.

05 / DASHBOARD

Dashboard design and layout

Layout schematic of the existing workbook, not a native Tableau screenshot.



Reading order

Start with the five headline measures. Compare geography and time trends, then examine seasonal volume, stay behaviour and destination rankings. Use the Tableau story to move through the additional contextual views.

Design rationale

- KPI cards provide a quick entry point before detailed analysis.
- Map, line, heatmap, scatter, bars and area views answer different questions.
- Visible State and Year controls support exploration, but their scope must be checked per worksheet.

Workbook inventory: 14 worksheets, one main dashboard and one story containing five points. The main dashboard uses automatic sizing.

06 / MEASURES

KPIs and their purpose

Verified all-data values; saved worksheet selections may show different values.

Workbook KPI	All-data value	Meaning
Total Tourist arrivals	210,311	SUM(Tourist_Arrival_Volume), from Tourist Data.
Total Tourism Revenue	5,730,625	SUM(Revenue); currency not specified.
Average Night Stayed	4.965 nights	AVG(Nights_Stayed) across 200 records; not volume-weighted.
Avg Revenue per Tourist	28,653.125	Revenue divided by 200 distinct Tourist_ID values.
No.of Active Hotels	40,081	SUM(Hotels_Available) over 150 observation rows; not unique hotels.
COVID Impact Score (story)	106.5 points	65 high x 1 + 70 moderate x 0.5 + 65 low x 0.1.

Two different arrival concepts

The dashboard arrival KPI totals 210,311 units in Tourist Data. Most map, destination and time charts instead use Tourism Growth Trends, whose all-data Tourist_Arrivals sum is 7,188,443.29. The two tables describe different observations and are not expected to reconcile.

Suggested presentation wording

“Average revenue per tourist” means revenue per distinct ID in this project. “Active hotels” is a sum of recorded hotel values; repeated observations can count the same capacity more than once.

07 / INTERACTION

Filters and story navigation

Explain the saved view separately from the full-data reference values.

Control / selection	Observed configuration
State control	Shown with the state-arrival map. The map has Madhya Pradesh selected in the saved workbook.
Year control	Associated with the cumulative Domestic/Foreign arrivals view.
Filter propagation	Global application to every worksheet is not established by the saved configuration. Check Apply to Worksheets in Tableau.
Top N selections	Destination chart uses Top 10 by arrivals. Revenue chart uses Top 5 by revenue. State-share pie uses a different ranking measure; see page 17.

Five story points

Order	Story content
1	COVID Impact Score On Tourism in India
2	See which Indian states attract the most tourists - state-share pie
3	Most contributed states - annotated state-share pie
4	Top 5 States by Revenue
5	Dashboard - complete overview

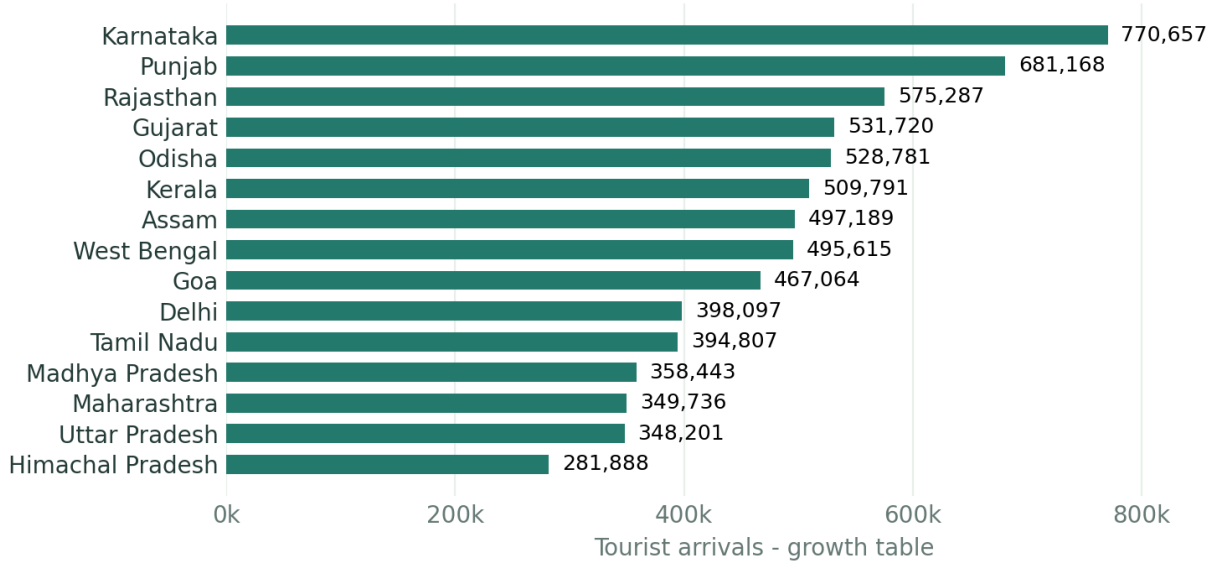
Presentation flow: open the dashboard, explain KPI definitions, inspect a state and a year range, then navigate through the story points. Restore filters before comparing with the all-data figures in this report. Story points may preserve their own saved selections.

08 / VISUALIZATION 01

State-wise tourist arrivals

Map in Tableau; all-state bars in this report

7,188,443 Total growth-table arrivals



WHAT DO YOU SEE?

Karnataka has the highest arrival total, 770,656.68, followed by Punjab at 681,167.61. The saved Tableau map selects Madhya Pradesh, whose total is 358,442.50.

POSITIVE SCENARIO

The map supports geographic comparison, while size and colour draw attention to states with higher recorded activity.

NEGATIVE SCENARIO

One selected state can hide the broader distribution. Raw totals also lack population or visitor-capacity normalization.

CONCLUSION

Karnataka leads within this extract, accounting for about 10.72% of the growth-table arrival total.

SUGGESTION

Clear the saved State selection for an overview and validate the source geography before making regional policy claims.

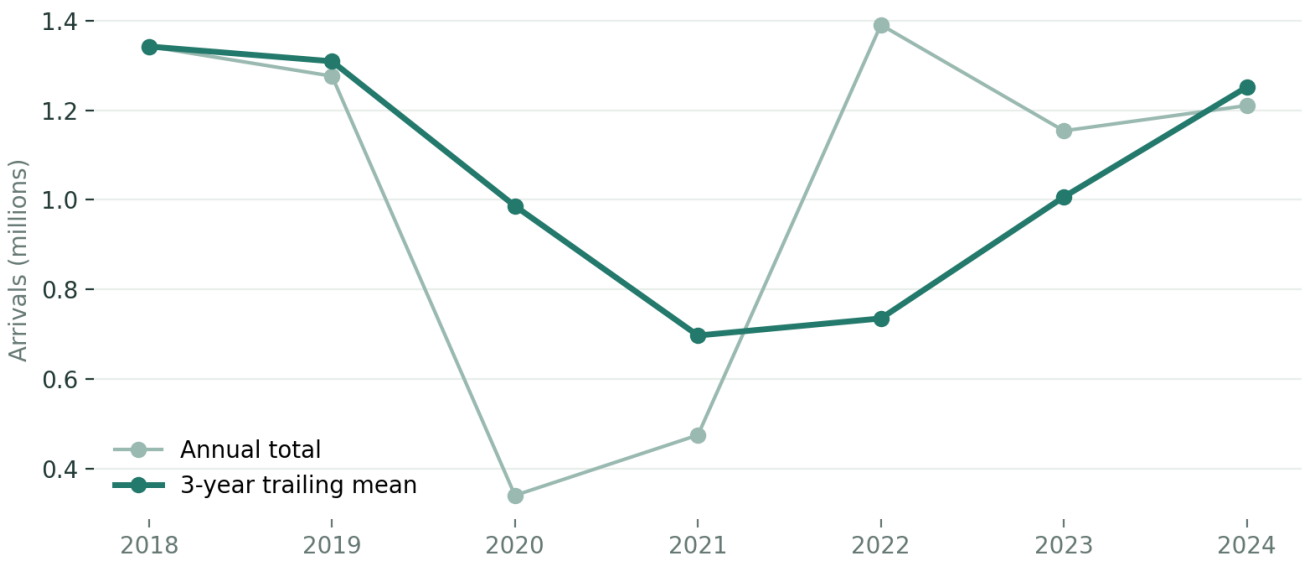
08 / VISUALIZATION 02

Moving average of arrivals

Annual totals and a trailing three-year mean, recreated

1,390,699

Highest annual total: 2022



WHAT DO YOU SEE?

Annual arrivals fall from 1,276,191 in 2019 to 340,018.30 in 2020, then recover to 1,390,699 in 2022. The trailing mean smooths these fluctuations.

POSITIVE SCENARIO

A moving average makes the broad direction easier to read when individual years vary sharply.

NEGATIVE SCENARIO

Smoothing delays turning points; the first two points use fewer than three years. The pattern alone does not prove a pandemic effect.

CONCLUSION

The extract shows a marked 2020 trough followed by recovery. The 2024 total, 1,210,453, remains below the 2022 peak.

SUGGESTION

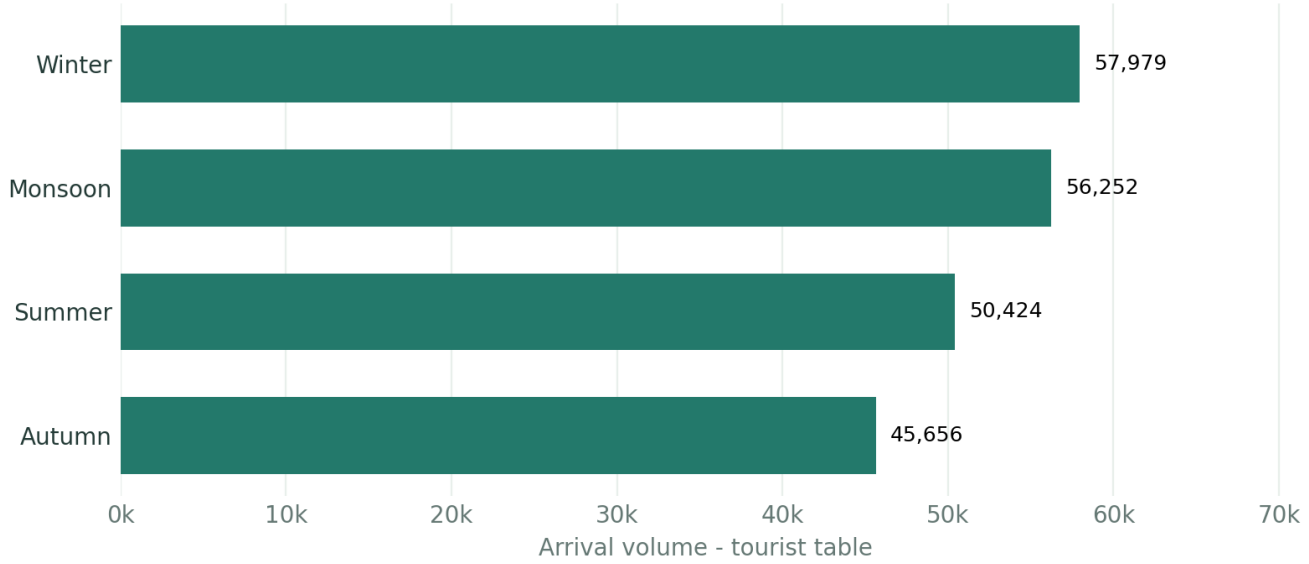
Explain the window as current year plus two preceding years and confirm that the calculation addresses Year in ascending order.

08 / VISUALIZATION 03

Season and year heatmap

Companion figure: season totals only; no invented annual split

57,979 Winter arrival volume



WHAT DO YOU SEE?

Winter contributes 57,979 arrival-volume units, Monsoon 56,252, Summer 50,424 and Autumn 45,656. The workbook arranges Season against Year as a heatmap.

POSITIVE SCENARIO

Colour intensity can make high and low seasonal volumes easy to compare when the underlying dates are valid.

NEGATIVE SCENARIO

Season is in Tourist Data, but Year is in another table related only by State. A cell does not establish visits in that season of that year.

CONCLUSION

The source supports an overall seasonal-volume comparison; the current relationship does not support a reliable year-season trend.

SUGGESTION

Add a visit date or year to Tourist Data before interpreting annual seasonality. Until then, present season totals independently.

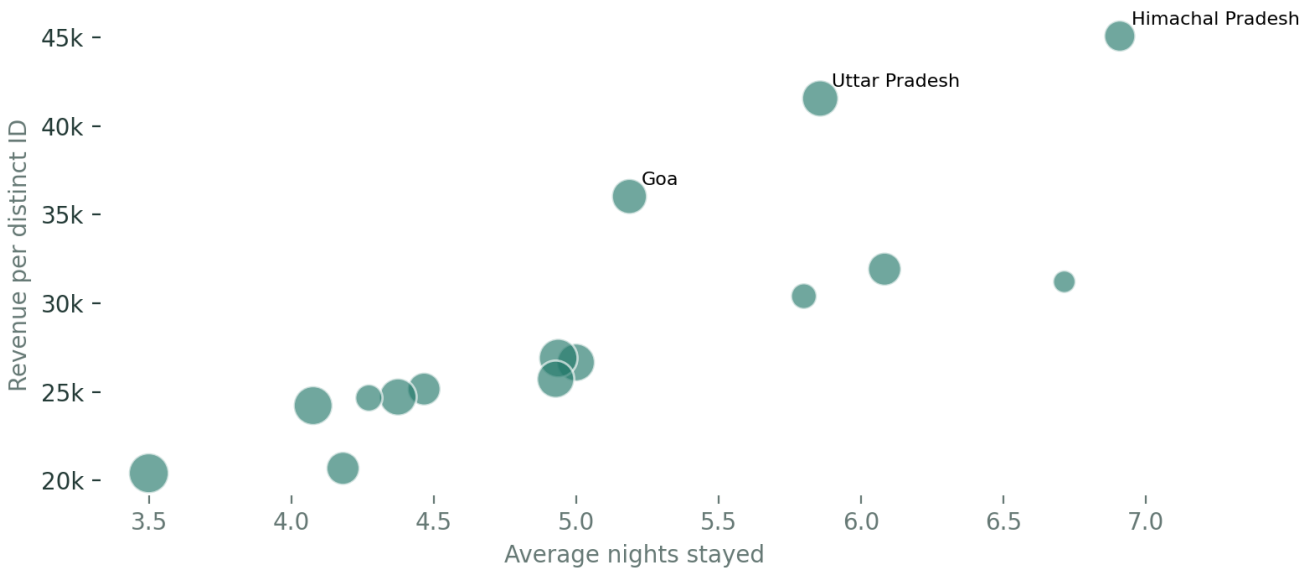
08 / VISUALIZATION 04

Stay duration and revenue

Companion scatter: one point per state, without Category colour

4.965

Average nights across tourist records



WHAT DO YOU SEE?

The chart relates average nights stayed to revenue per distinct tourist ID, with arrival volume controlling bubble size. The report aggregates 15 states using the ID-matched tables.

POSITIVE SCENARIO

This view helps compare spending and stay patterns in a single display and identify states for closer investigation.

NEGATIVE SCENARIO

Association is not causation. The workbook colours by a category from the state-related growth table, which does not identify individual visitors as Domestic or Foreign.

CONCLUSION

The visitor and revenue tables support state-level comparisons; the chart does not prove that longer stays cause higher revenue.

SUGGESTION

Use visitor-level category data before comparing Domestic and Foreign spending. Explain bubble size separately from the revenue axis.

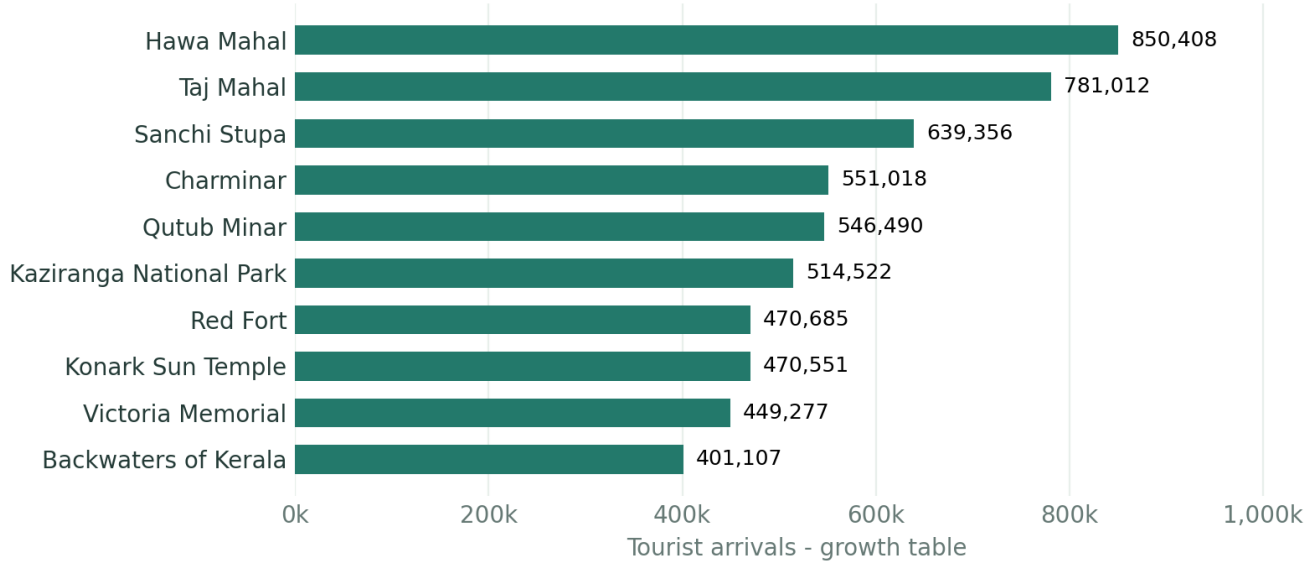
08 / VISUALIZATION 05

Top ten destinations

Horizontal bars: SUM(Tourist_Arrivals), sorted descending

850,408

Hawa Mahal arrivals, rounded



WHAT DO YOU SEE?

Hawa Mahal leads with 850,407.75 arrivals, followed by Taj Mahal with 781,012.27 and Sanchi Stupa with 639,355.78.

POSITIVE SCENARIO

Sorted bars make the destination ranking and differences between totals easy to understand.

NEGATIVE SCENARIO

A Top 10 view hides the remaining five destinations. Some state-destination associations in the source require correction.

CONCLUSION

These are leading destination labels within the supplied observations, not a verified national ranking of monuments.

SUGGESTION

Validate place-to-state mappings, keep the Top 10 measure consistent, and state the selected year range when presenting a filtered ranking.

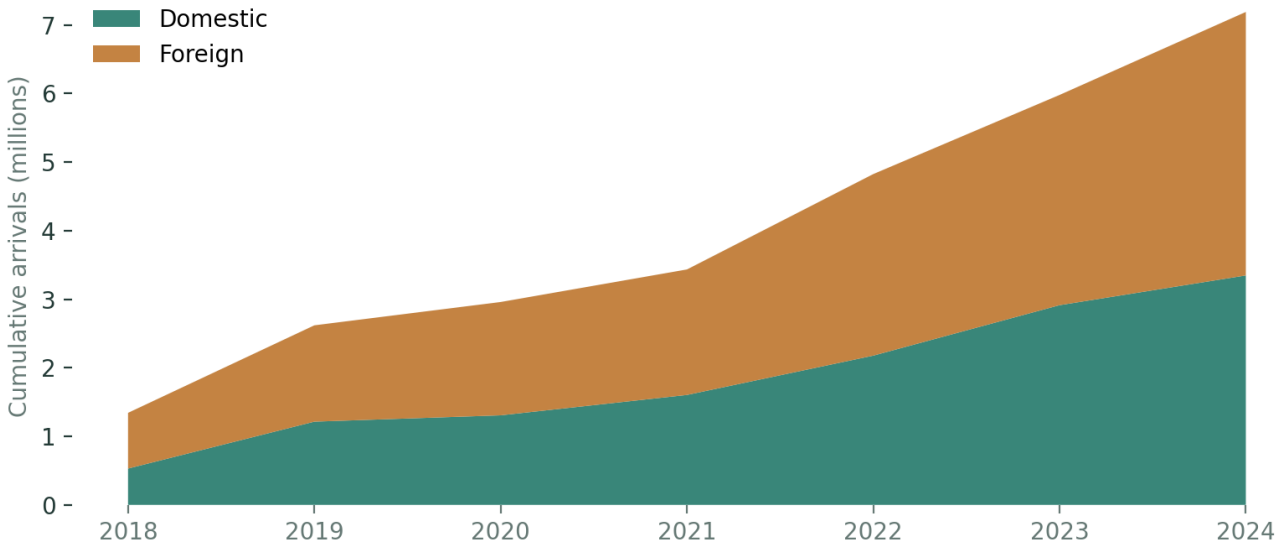
08 / VISUALIZATION 06

Domestic and foreign arrivals

Stacked area with cumulative totals by Year

53.46%

Foreign share of all growth-table arrivals



WHAT DO YOU SEE?

By 2024, cumulative Foreign arrivals total 3,842,605.52 and Domestic arrivals total 3,345,837.77, together 7,188,443.29.

POSITIVE SCENARIO

The stacked area shows how cumulative totals accumulate and how the two categories contribute to the combined sum.

NEGATIVE SCENARIO

A running total generally rises even when annual arrivals fall. Reading its slope or height as a single-year total would be misleading.

CONCLUSION

Foreign observations contribute slightly more of the total in this extract. This should not be generalized to all tourism in India.

SUGGESTION

Call the view “Cumulative arrivals by category” and pair it with annual values when discussing the 2020 decline or recovery.

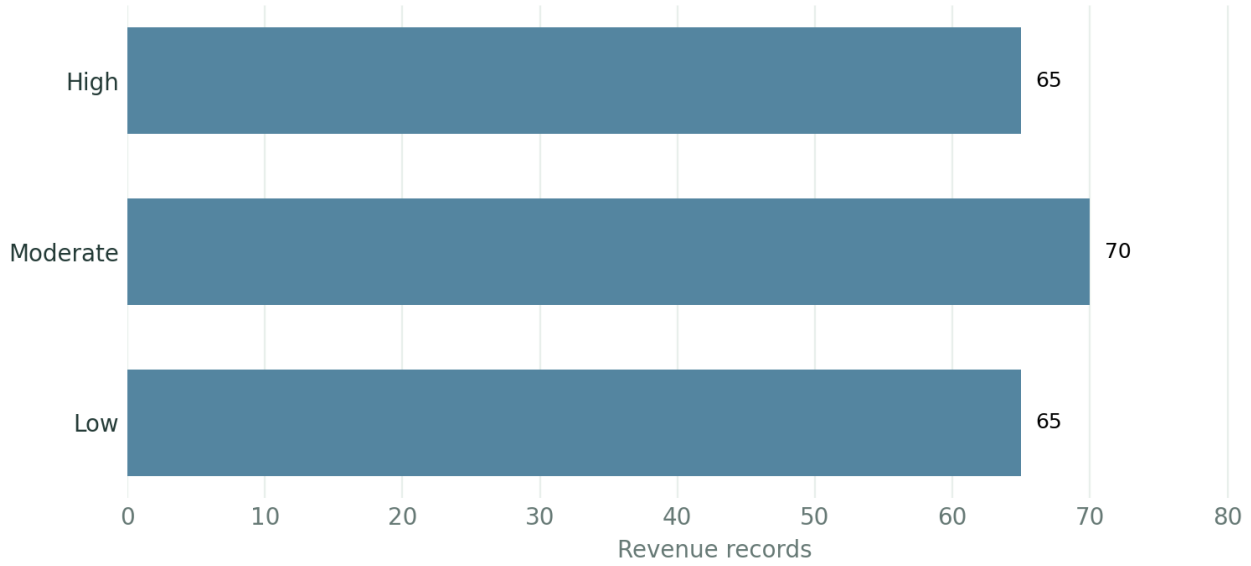
08 / VISUALIZATION 07

COVID-impact score

Companion bars: category counts behind the story KPI

106.5

Weighted score points



WHAT DO YOU SEE?

The Revenue table contains 65 High, 70 Moderate and 65 Low records. Applying weights of 1, 0.5 and 0.1 produces the story score of 106.5.

POSITIVE SCENARIO

The score summarizes labelled impact categories in one value, while the count chart reveals its composition.

NEGATIVE SCENARIO

The weights are a project assumption and the sum grows with record count. It is neither a percentage nor a measured financial loss.

CONCLUSION

This is a descriptive weighted index of source labels, not evidence of the causal effect of COVID on tourism.

SUGGESTION

Always show the weights and sample size. A mean score of 0.5325 can support like-for-like comparisons, if clearly labelled as a proposed measure.

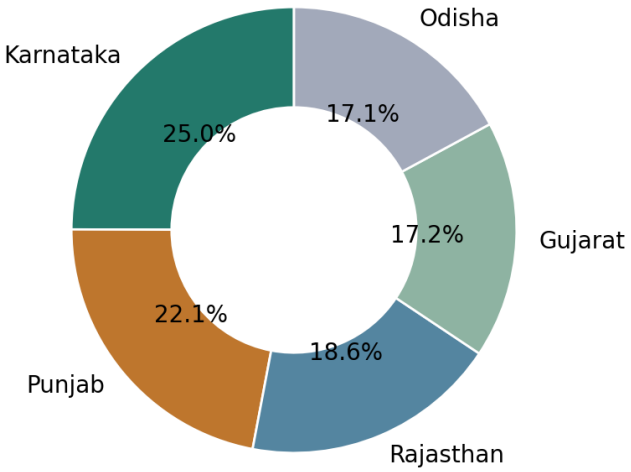
08 / VISUALIZATION 08

State share of arrivals

Companion: top five by arrivals, shares within those five

42.95%

Top five arrivals as a share of all states



WHAT DO YOU SEE?

Using a consistent arrival measure, the top five are Karnataka, Punjab, Rajasthan, Gujarat and Odisha, together 3,087,611.90 arrivals. The companion donut shows their within-group shares.

POSITIVE SCENARIO

A share chart communicates contribution within a clearly defined set of states.

NEGATIVE SCENARIO

The saved pie mixes measures: wedge size uses arrivals, Top 5 ranks COUNT(Category), and labels use percent of COUNT(State). Its percentages do not describe arrival share.

CONCLUSION

The original pie cannot support an arrival-share claim from its labels. The companion figure is a separate, consistent calculation and does not reproduce that saved pie.

SUGGESTION

Use SUM(Tourist_Arrivals) for ranking, wedge size and percentage labels, then explicitly choose an all-state or displayed-top-five denominator.

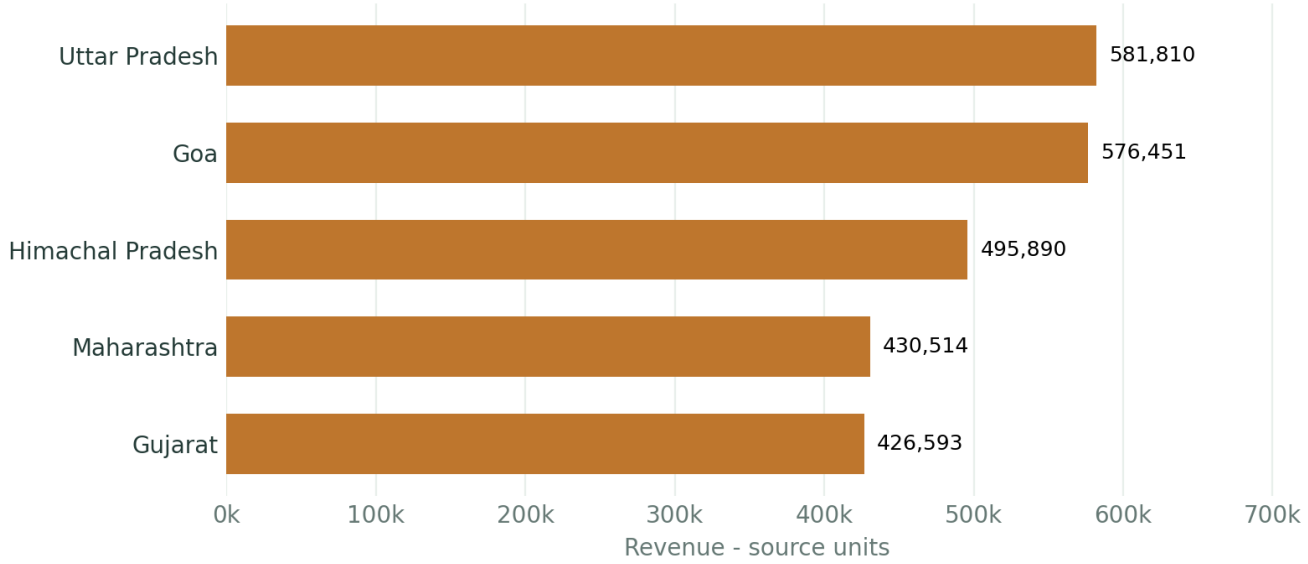
08 / VISUALIZATION 09

Top five states by revenue

Revenue totals by state; all-data ranking

581,810

Uttar Pradesh revenue, source units



WHAT DO YOU SEE?

Uttar Pradesh leads at 581,810, followed by Goa at 576,451, Himachal Pradesh at 495,890, Maharashtra at 430,514 and Gujarat at 426,593.

POSITIVE SCENARIO

The ranking highlights the states contributing most to the project revenue total and supports focused comparison.

NEGATIVE SCENARIO

Revenue has no year field. A statement about revenue specifically in 2024 is not established by this table; currency is also unspecified.

CONCLUSION

The five states contribute 2,511,258, or about 43.82% of all recorded revenue. Goa and Karnataka together contribute about 16.74%, without a year attribution.

SUGGESTION

Use a dated revenue source for yearly claims. Explain these values as source units and revise unsupported story annotations before presentation.

09 / FINDINGS

Conclusion and recommendations

A useful exploratory dashboard, with clear measurement boundaries.

Main findings

- Karnataka leads recorded growth-table arrivals; Uttar Pradesh leads the Revenue table. Arrival leadership and revenue leadership are different questions.
- The annual arrival series has a 2020 trough and a 2022 peak. A moving average smooths this profile, while the area chart accumulates it.
- Winter has the highest overall tourist-table arrival volume, though seasonal observations are not dated by year.
- Foreign category observations contribute 53.46% of growth-table arrivals. The extract does not establish national market shares.

Recommendations

Priority	Action
Before presenting	Explain the two arrival measures, the running total and the hotel-sum definition.
Before making share claims	Align ranking, slices and labels in the state-share pie; remove unsupported 2024 revenue statements.
For stronger analysis	Obtain dated visitor and revenue data, verified destination geography, currency and source methodology.
For better usability	Confirm State/Year filter scope and test saved story selections in Tableau.

Project value: the workbook demonstrates multiple Tableau techniques and a coherent tourism theme. Its strongest conclusions are descriptive comparisons within the supplied data, with explicit field definitions and limitations.

10 / CALCULATIONS

Calculated fields and KPI formulas

Existing formulas, with logical table context made explicit.

Average revenue per tourist - existing calculated field

```
SUM([Revenue]) / COUNTD([Tourist_ID])
```

Denominator: 200 distinct IDs at all-data scope. This differs from dividing by either arrival-volume field.

COVID score - existing row-level calculated field

```
IF [COVID_Impact] = "High" THEN 1
ELSEIF [COVID_Impact] = "Moderate" THEN 0.5
ELSE 0.1
END
```

COVID_Impact here belongs to Revenue. The story aggregates this field with SUM. ELSE also assigns 0.1 to unexpected values, so future data should be validated.

KPI / measure	Expression
Arrival KPI	SUM([Tourist_Arrival_Volume])
Revenue KPI	SUM([Revenue])
Average nights	AVG([Nights_Stayed])
Hotel KPI	SUM([Hotels_Available])
Growth-table arrivals	SUM([Tourist_Arrivals])
COVID story KPI	SUM([Covid_Score])

Field names in this report use their readable source names. Tableau may display spaces or table suffixes to distinguish duplicate names such as State and COVID Impact.

11 / CALCULATIONS

Time calculations and ranking

Calculation direction, filtering and denominator determine the meaning.

Trailing moving average - equivalent expression

```
WINDOW_AVG(SUM([Tourist_Arrivals]), -2, 0)
```

Address Year in ascending order, within each intended partition. The current year and previous two available year marks form the window. The first mark uses one year and the second uses two. The workbook saves an average window from -2 to 0.

Cumulative category arrivals - equivalent expression

```
RUNNING_SUM(SUM([Tourist_Arrivals]))
```

Compute along Year and restart for each Category. Filtering out earlier years may change the start of the running sum; explain the displayed period.

Consistent share - proposed correction, not already applied

```
SUM([Tourist_Arrivals])
/ TOTAL(SUM([Tourist_Arrivals]))
```

Compute across State. After a Top 5 filter this normally describes the displayed five states; it does not automatically use the all-state denominator. For all-state share, preserve a correctly scoped total and test filter order.

Ranking	Existing selection
Destinations	Top 10 by SUM(Tourist_Arrivals)
Revenue states	Top 5 by SUM(Revenue)
State-share pie	Top 5 by COUNT(Category); different from arrival ranking

12 / IMPLEMENTATION

Build procedure

A reproducible sequence for explaining how the Tableau project is assembled.

1

Connect the data

Open the packaged workbook. The embedded Hyper extract provides the three logical tables; use the original Excel file only if available.

2

Verify relationships

Relate Tourist Data to Revenue on Tourist_ID and Tourist Data to Tourism Growth Trends on State. Confirm key types and table grain.

3

Create calculated fields

Add the average-revenue formula and COVID-score mapping. Keep aggregation separate from the row-level score.

4

Create KPI worksheets

Use text marks for arrival volume, revenue, average nights, revenue per distinct ID and hotel-value sum.

5

Build the six dashboard views

Configure the map, moving average, season-year heatmap, stay/revenue scatter, Top 10 destination bars and cumulative category area chart.

6

Assemble the dashboard

Arrange KPI cards above the analytical views, apply readable number formatting, and show the State and Year controls with deliberate scope.

7

Build the story

Add the COVID score, state-share view, annotated state-share point, revenue ranking and dashboard overview in the saved order.

8

Validate and package

Check totals, calculation direction, labels and filter behaviour. Save as TWBX to include the extract and required assets.

13 / EXPLANATION

Presentation and viva guide

A concise speaking structure supported by the verified project facts.

Suggested opening

“Our project explores tourism trends in India using three related tables. The dashboard summarizes arrival volume, revenue and visitor stays, while the story explains selected state patterns and a weighted COVID-impact index. All results refer to the supplied project extract.”

Questions you should be ready to answer

Question	Suggested answer
Why are there two arrival totals?	They come from different tables and grains: 210,311 in Tourist Data and 7,188,443.29 in Tourism Growth Trends.
What does the moving average do?	It averages the current and two preceding annual marks to smooth short-term fluctuations.
Why does the area chart keep rising?	It displays running totals, so it accumulates arrivals rather than showing each year independently.
Does longer stay cause more revenue?	The scatter is exploratory. Correlation and state-level averages cannot establish causation.
Is the COVID score a percentage?	No. It is a weighted sum of categorical labels using 1, 0.5 and 0.1.
Are the hotel values unique hotels?	No. The KPI sums recorded Hotels_Available values across observation rows.
Can the heatmap prove annual seasonality?	Not with the current State-only relationship; visit-year data is needed.

Suggested close: highlight the leading states and destinations, explain the time pattern, then acknowledge the source and relationship limitations. Keep recommendations proportional to the available data.

14 / REFERENCE

Sources and verification

Traceable to the supplied workbook and its embedded data.

Primary project materials

- Packaged workbook: Tourism_Trends_India_Dashboard(1).twbx.
- Workbook definition: Tourism_Trends_India_Dashboard.twb inside the package.
- Embedded data: packaged Hyper extract with Tourist Data, Revenue and Tourism Growth Trends.
- Original source caption: Tourist_Growth_Trends_India_Final; upstream source unverified.

What was verified

- Table sizes, field names, null counts, identifier uniqueness and state coverage.
- KPI totals and the report charts calculated from the embedded tables.
- Worksheet inventory, saved filter configuration, calculated fields, table-calculation settings and story-point sequence.

Figure and interpretation notes

All report charts were independently rendered from the embedded extract. The dashboard diagram is schematic. The state bars, seasonal bars, state-level scatter, COVID-category bars and consistent top-five donut are explanatory companion views, not screenshots of the existing Tableau worksheets.

Workbook structure was inspected without changing the submitted file. Native Tableau rendering, tooltip behaviour and live filter propagation were not tested. Proposed corrections in this report are recommendations, not completed workbook edits.

Responsible conclusion

Use this as an academic analysis of the supplied extract. Official tourism, revenue or infrastructure claims require a verified source, consistent geography, clear units and dated observations.